

Dual Tree Complex Wavelet Transform Domain Based Motor Imagery Movements Classification Scheme from EEG Signals

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Abstract—In this paper, a feature extraction method based on Dual Tree Complex Wavelet Transform (DTCWT) domain has been proposed to classify left and right hand motor imagery movements from electroencephalogram (EEG) signals. After first performing auto-correlation of the EEG signals to reduce noise and enhance the weak brain signals, the EEG signals are decomposed into several bands of real and imaginary coefficients using DTCWT. Then mean absolute deviation has been extracted as features from relevant coefficient bands and from the one way ANOVA analysis, scatter plots, box plots and histograms, this feature has shown to be promising to distinguish various kinds of EEG signals. The total experiment is carried out with the publicly available benchmark BCI-competition 2003 Graz motor imagery data set. Among different types of classifiers used K-nearest neighbor (KNN)-based classifiers have been shown to provide a good mean accuracy of 90.36% which is better than several existing techniques.

Keywords—*Electroencephalogram (EEG); BCI; Dual Tree; Complex Wavelet Transform (DTCWT); Auto correlation; Mean absolute deviation; KNN classifier*

I. INTRODUCTION

Brain computer interfacing (BCI) allows controlling and operation of computer aided systems by intent alone. The major objective of BCI system is to assist disable people for their rehabilitation. BCI involves detection, analysis and classification of different types of motor imagery movements to implement real time control and communication. Electroencephalogram (EEG) signals are often used for BCI purpose since it can be implemented as a non-invasive system. One major category of BCI is the detection of motor imagery movements such as left and right hand movements. Many methods have been developed so far in the literature for classifying various types of arm movements. Separability of EEG signals using adaptive auto regressive parameter based method was discussed in [1] and [2]. In [3], wavelet coefficients, power spectral density and average power obtained from EEG signals have been used as features to classify left and right hand movements with different classifiers such as linear discriminant analysis (LDA), support vector machine (SVM) and KNN. In [4], wavelet packet entropy of EEG signal is used as feature and SVM as the classifier.

Since EEG data for research purpose can be acquired with varying experimental setup and conditions, BCI competition was held providing standard data sets to evaluate and compare

different algorithms. Different approaches have been studied to classify motor imagery movements in BCI competition 2003 Graz motor imagery data set. Band passed EEG signals and power spectral density based LDA was described in [5] while in [6], raw EEG signal incorporated with Hidden Markov Model (HMM) was presented by the same author. Adaptive Auto Regressive (AAR) model based features with Bayesian Graphical Network (BGN) and Multi Layer Perceptron were reported in [7]. Morlet wavelet was used to extract features from mu rhythms with Bayes quadratic classifiers in [8]. Recently, multiple auto-correlation based feature extraction method along with learning vector quantization (LVQ) has been proposed in [9].

The objective of this paper is the identification of imagery hand movements by extracting suitable features from pre-processed EEG signal in the dual tree complex wavelet transform (DTCWT) domain. DTCWT has been widely used for image and video processing [10], and recently, it has been used in the area of bio-medical signals [11] - [13]. Since motor imagery movements occur in low frequency EEG bands [14], to perform detail analysis on specific bands, DTCWT has been used as it is a rich level analysis tool than the traditional discrete wavelet transform (DWT). First auto correlation has been performed on the EEG signal to enhance it and reduce the effect of noise. After that mean absolute deviation is extracted as features from DTCWT real and imaginary coefficients of the auto-correlated EEG signal and the discrimination capability of the selected feature has been analyzed through various types of scatter plots, box plots, histograms and one way ANOVA analysis. Finally, KNN classifier has been employed for the purpose of classification.

II. DESCRIPTION OF THE EEG DATABASE

BCI competition 2003 data set (motor imagery III) provided by Technical University of Graz has been used in this paper. This data was acquired from a normal subject sitting in a chair with arm rests. Subject was trying to control a feedback bar by making imagery movements of left or right hands. Left and right cues were in random order [15]. The experiment had of 7 runs with 40 trials each. During each trial, the beginning was indicated at $t = 2s$ an acoustic stimulus and a cross '+' was displayed for 1s. After this an arrow (left or right) was displayed as the cue at $t = 3s$. At the same time the subject was asked to move a bar into the direction of the cue which was controlled by adaptive auto-regressive parameters of channel C3 and C4. The EEG signal was filtered between

0.5 and 30 Hz with the sampling rate of 128 Hz. A detail description of the experimental set up can be found in [16].

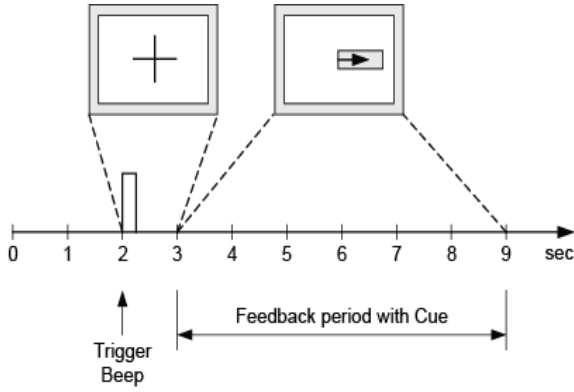


Fig. 1. Timing scheme of the experiment.

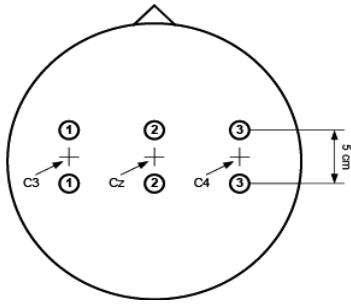


Fig. 2. Electrode positions.

III. DUAL TREE COMPLEX WAVELET TRANSFORM

Dual tree complex wavelet transform (DTCWT) is a recent enhancement to the discrete wavelet transform (DWT) which has additional properties including nearly shift invariant and directionally selective in two and higher dimensions [17]. DTCWT is 2^d times redundant for any d dimensional signal as compared to DWT and offers directional information in six directions. Thus is it more efficient in time frequency localization of EEG signal.

Similar to positive/negative post-filtering of real sub-band signals, the idea behind dual tree approach is quite simple. DTCWT employs two real DWTs where the first DWT gives the real part of the transform while the second DWT gives the imaginary part. The analysis filter bank structure used to implement DTCWT is given in Fig. 3. Two real wavelet transforms use two different sets of filters which satisfy perfect reconstruction conditions.

If square matrices $\mathbf{H}_h$ and $\mathbf{H}_g$ denote the two real DWTs, then the DTCWT can be represented as follows:

$$\mathbf{H} = \begin{bmatrix} \mathbf{H}_h \\ \mathbf{H}_g \end{bmatrix} \quad (1)$$

The inverse transform of $\mathbf{H}$ is given as,

$$\mathbf{H}^{-1} = \frac{1}{2} \begin{bmatrix} \mathbf{H}_h^{-1} & \mathbf{H}_g^{-1} \end{bmatrix} \quad (2)$$

If the vector $\mathbf{x}$ represents a real signal, then $\mathbf{w}_h = \mathbf{H}_h \mathbf{x}$ represents the real part and $\mathbf{w}_g = \mathbf{H}_g \mathbf{x}$ represents the imaginary part of the DTCWT.

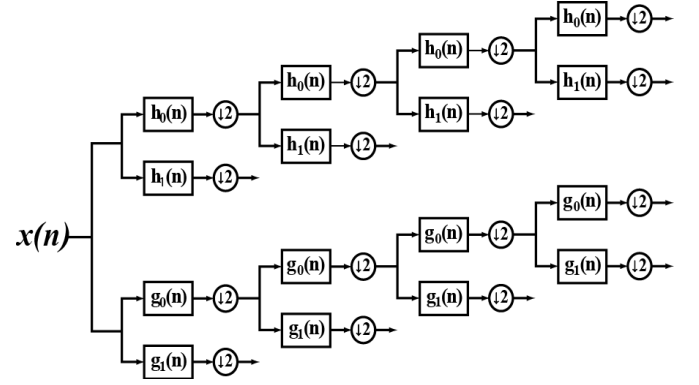


Fig. 3. Analysis filter bank of DTCWT.

When the dual-tree CWT is applied to a real signal, the output of the upper and lower filter banks gives the real and imaginary parts of the complex coefficients respectively. Among several available wavelets, Farras [18] wavelets are used to perform DTCWT.

IV. ANALYSIS IN DUAL TREE COMPLEX WAVELET TRANSFORM DOMAIN

The proposed method consists of first auto-correlating the EEG signals to attenuate noise effects; secondly decomposing the EEG signal of each trial with dual tree complex wavelet transform into three levels and finally extracting suitable features from different bands. After extracting suitable features, KNN classifier is deployed for classifying hand movements.

A. Feature Extraction in DTCWT Domain

Since BCI 2003 Graz data set was recorded with low pass filtering of 30 Hz, no preprocessing was needed to discard the unnecessary high frequency components. However, since EEG signal is very weak and corrupted with noise, auto-correlation has been performed on each trial of EEG signals so that we can analyze the desired low passed signal in noise reduced space. Auto correlation enhances the EEG signal [9] and makes it suitable for analysis. Since motor imagery activity occurs in low frequency EEG band signal [14], so to have in-depth view, dual tree complex wavelet transform is applied to the acquired EEG signal. Since forward transform of DTCWT gives two branches containing real and imaginary coefficients, it gives a rich way of analyzing EEG signal than discrete wavelet transform (DWT). The experiment was carried out by taking feedback from C3 and C4 channel while making Cz the reference. As a result, for applying DTCWT, EEG signal acquired from both C3 and C4 channel for a single trial is decomposed into three levels. If the original low passed signal is denoted by X which has 0.5 to 30 Hz frequency components, after first level decomposition, it provides Y_1 (16-30 Hz) and Z_1 (0.5-15 Hz). After second level decomposition, Z_1 leads to Y_2 (7.5-15 Hz) and Z_2 (0.5-7.5 Hz). So after three levels of DTCWT, the four frequency bands are Y_1 (16-30 Hz), Y_2 (7.5-15 Hz), Y_3 (3.75-7.5 Hz) and Z_3 (0.5-3.75 Hz).

Hz). Reconstructions of these components using the inverse DTCWT approximately correspond to the physiological EEG sub-bands delta, theta, alpha and beta respectively [19]. Since each frequency band gives both real and imaginary coefficients, we have total 4 real and 4 imaginary coefficient bands. From now on, the real and imaginary bands and associated levels will be denoted as RB_x and IB_x where 'x' is the level index. Fig. 4 represents the recorded EEG signals from C3 channel for left and right hand imagery movements and the auto-correlated signal. Fig. 5 shows the corresponding real and imaginary coefficients of the first and second level DTCWT.

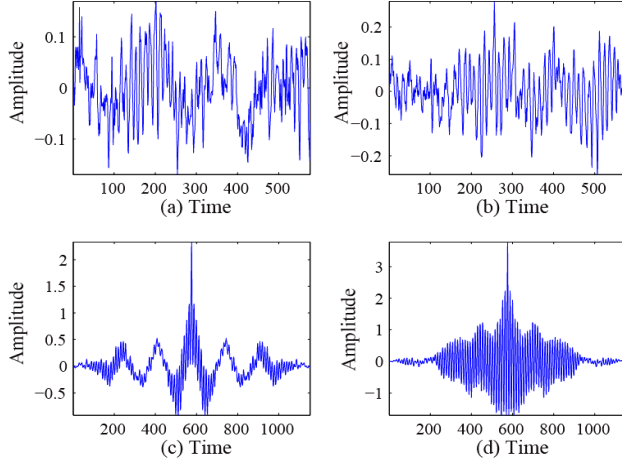


Fig. 4. (a)-(b) EEG signal from C3 channels for left and right hand movement; (c)-(d) auto correlated signal of (a) and (b) respectively.

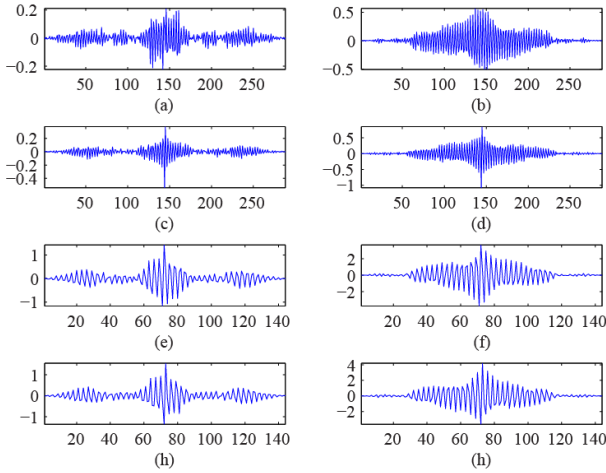


Fig. 5. (a)-(b) Real coefficients of level 1 for left and right imagery movements respectively; (c)-(d) imaginary coef. of level 1 for left and right movements respectively; (e)-(f) real coef. of level 2 for left and right movements respectively; (g)-(h) imaginary coef. of level 2 for left and right imagery movements respectively.

The underlying dynamics of EEG signals is spread over various sub-bands in the frequency domain. To classify motor imagery movements, we need to extract more information in lower frequency bands of EEG signals and mu rhythms (7.5-12.5 Hz) [8]. For this purpose, mean absolute deviation in different bands have been extracted as features.

Mean Absolute Deviation (MAD) of a data set is the average distance between each data value and the mean value. It measures the variability of a univariate sample of quantitative data or simply it is a summary of statistical dispersion.

$$MA \quad (3)$$

where N denotes the number of data points, x_i is the i -th sample of the data set, $\bar{x}$ is the mean of N samples calculated as:

$$\bar{x} = \frac{1}{N} \sum_{i=1}^N x_i \quad (4)$$

Table I gives the p-values of one way ANOVA analysis of MAD for different bands. The hypothesis about the p-values is that the value $p < 0.05$ indicates that at least one sample mean is significantly different than the other sample means statistically [20]. From Table I, it is clear that MAD values of 2nd and 3rd level coefficients have very small p-values since these corresponds to alpha wave and mu rhythms of EEG signals which differ during motor imagery hand movements. So MAD of these two bands can be used as good features.

TABLE I. RESULTS OF ANOVA ANALYSIS

Feature bands	p-value from ANOVA
$RB_2(C3)$	1.2180e-19
$IB_2(C3)$	1.2181e-08
$RB_3(C3)$	1.9784e-20
$IB_3(C3)$	1.5199e-08
$RB_2(C4)$	1.0588e-07
$IB_2(C4)$	1.0935e-07
$RB_3(C4)$	5.2407e-08
$IB_3(C4)$	5.7204e-08

Apart from ANOVA analysis, scatter plots, box plots and histograms are provided to further illustrate the classification capability of the features.

Fig. 6 represents the scatter plots of RB_2 and IB_2 for C3 channel respectively for MAD whereas Fig. 7 presents the scatter plots of RB_3 and IB_3 for C3 respectively. In Fig. 6 and Fig. 7, blue hexagons and red circles indicate feature values during left and right hand imagery movements respectively. From the figures it can be noted that the blue and red markers have significantly different values indicating their variation during different imagery hand movements and as a result, these can be used as good features.

Fig. 8 shows the box plots of $RB_2(C3)$ and $IB_2(C3)$, respectively for left and right hand imagery movements. The box plots indicate that the features have distinct values for the two specific movements and as a result they can be used in classifiers.

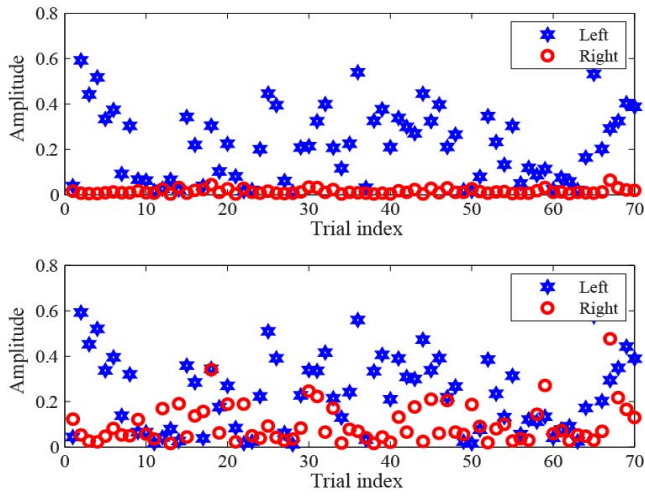


Fig. 6. (a)-(b) Scatter plots of RB_2 and IB_2 for MAD during left and right hand imagery movements respectively (in C3 channel).

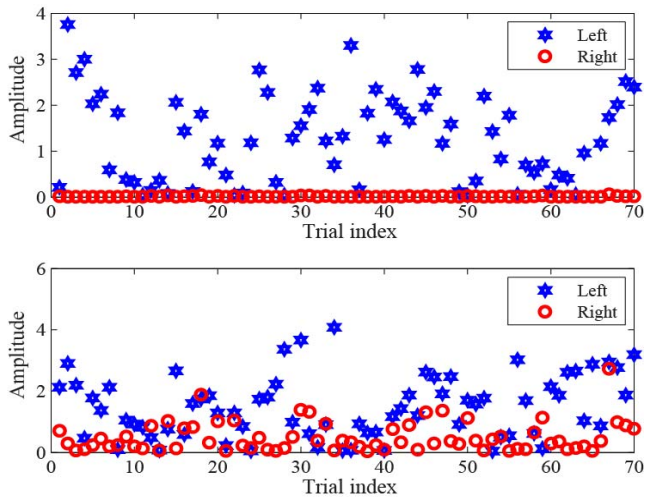


Fig. 7. (a)-(b) Scatter plots of RB_3 and IB_3 for MAD during left and right hand imagery movements respectively (in C3 channel).

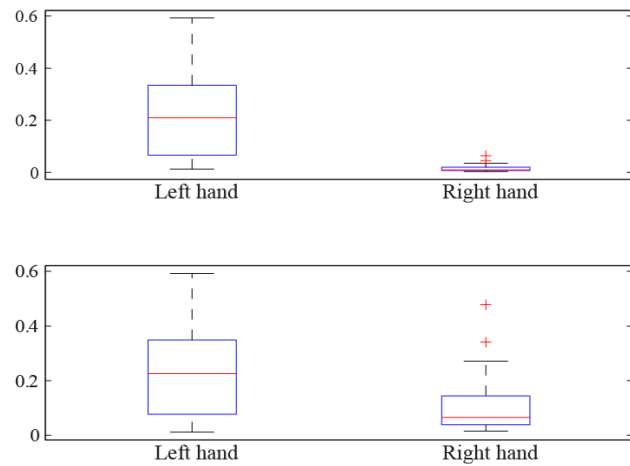


Fig. 8. (a)-(b) Box plots of $RB_2(C3)$ for left and right hand imagery movements respectively; (c)-(d) box plots of $IB_2(C3)$ for left and right hand imagery movements respectively.

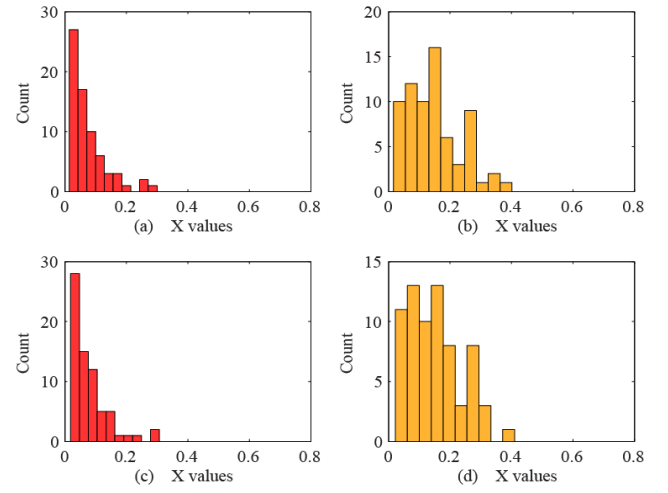


Fig. 9. (a)-(b) Histograms of $RB_2(C4)$ for left and right hand imagery movements respectively; (c)-(d) histograms of $IB_2(C4)$ for left and right hand imagery movements respectively.

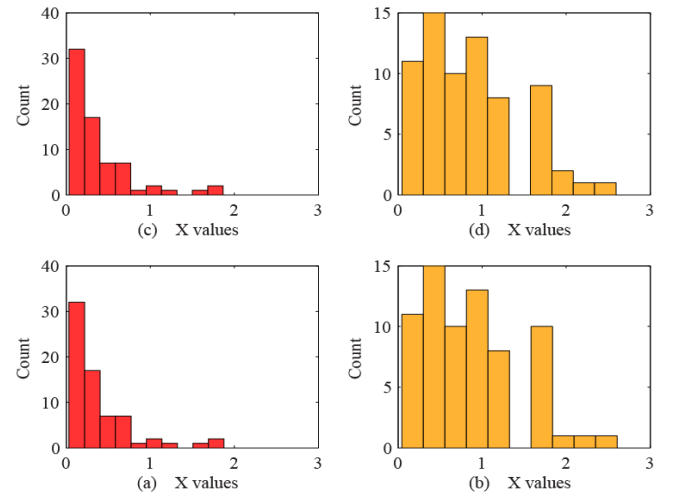


Fig. 10. (a)-(b) Histograms of $RB_3(C4)$ for left and right hand imagery movements respectively; (c)-(d) histograms of $IB_3(C4)$ for left and right hand imagery movements respectively.

Fig. 9 and Fig. 10 present the histograms of $RB_2(C4)$, $IB_2(C4)$, $RB_3(C4)$ and $IB_3(C4)$ for MAD where red and yellow color indicates left and right hand imagery movements respectively. From the histograms, it can be said that the corresponding histograms have peaks in different regions which is the evidence that the features show different values during two specified motor imagery movements. For example, $IB_2(C4)$ has peak in 0.04-0.05 region for left hand imagery movement while for right hand imagery movements, it has some sort of uniform distribution in 0-0.10.

The scatter plots, box plots, histograms and the p-values of one way ANOVA represent that MAD of real and imaginary coefficients from different bands of DTCWT of low frequency EEG signals have distinguishable values for left and right hand motor imagery movements. In other words, the features have good between-class distance and small within-class variance in the feature vector space [21] and as a result they can be used as suitable features.

B. Classification Using KNN Classifier

For any classification problem, there are two main parts - first is feature extraction and second is classification using the extracted features. If suitable features can be extracted, then a simple classifier can provide the desired outcome. Among different classifiers, KNN classifier has served best in our study.

K-Nearest Neighbors algorithm (KNN) is a non-parametric learning algorithm method used for classification. Among the various methods of supervised statistical pattern recognition, the Nearest Neighbor rule achieves consistently high performance, without a priori assumptions about the distributions from which the training examples are drawn. A new sample is classified by calculating the distance to the nearest training case; the sign of that point then determines the classification of the sample. The K-NN classifier extends this idea by taking the k nearest points and assigning the sign of the majority [22]. The number “ K ” decides how many neighbors influence the classification. If $k = 1$, then the algorithm is simply called the nearest neighbor algorithm.

V. PROPOSED METHOD AND RESULTS

Since it has been shown that mean absolute deviation varies significantly in different sub-bands of DTCWT of auto-correlated EEG signals, the proposed method extracts MAD as feature from RB_2 , IB_2 , RB_3 and IB_3 bands for both C3 and C4 channels. So, the feature vector (FV) of total length eight can be expressed as follows where T denotes transpose operation:

$$FV = [RB_2(C3), IB_2(C3), RB_3(C3), IB_3(C3), RB_2(C4), IB_2(C4), RB_3(C4), IB_3(C4)]^T \quad (5)$$

The EEG data set has 140 trials each for left and right hand (total 280 trials) of 9 seconds length. Since the cue was given at $t = 3$ sec, data segment after 3 seconds from C3 and C4 channels are used for classification. We have used four features from each channel, so a total of eight features from C3 and C4 channels form the final feature vector. The train and test feature matrix has dimension of 140×8 which is fed to KNN classifier. It is to be noted that we have used both training and testing data set for our experiment. The experiment was carried out using MATLAB 2013b [20] on Windows-7 32 bit platform having 1 GB RAM and 2.93 GHz Intel Core 2 Duo processor. For validation leave one out method (one-fold cross validation) is used.

KNN classifier has two parameters to tune- distance parameter and “ K ” value. The distance parameter is selected as “cosine” which means one minus the cosine of the included angle between points [20]. Another parameter of KNN classifier is the value of “ K ” which is task dependent [23]. For choosing the optimum “ K ” value, we have varied the value of “ K ”. Fig. 11 shows the variation of “mean accuracy (%)” for different “ K ” values and from this figure it is clear that for $K = 9, 11, 13, 14$ the classifier provides highest mean accuracy 90.36%.

Table II provides the performance evaluation of different classifiers which indicates that KNN classifier with “cosine” distance gives better mean accuracy than other different classifiers with various parameters for our problem.

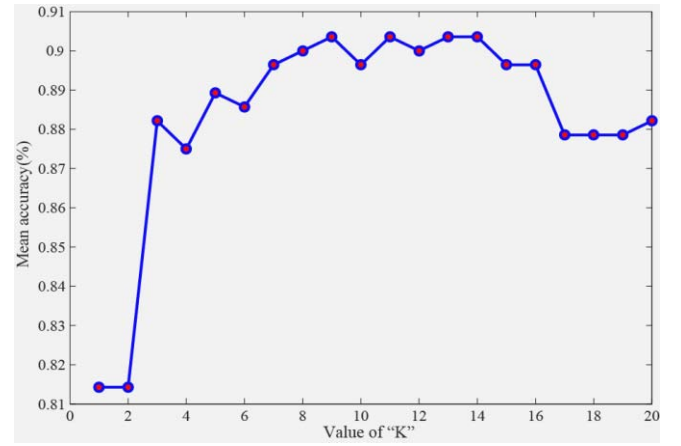


Fig. 11. Mean accuracy (%) vs different “ K ” values.

TABLE II. RESULTS OF ANOVA ANALYSIS

Classifier name	Classifier parameter	Mean Accuracy (%)
Naïve Bayes	Gaussian Distribution	84.64
Discriminant Analysis	Quadratic	81.07
	Mahalanobis	84.64
	Linear	86.43
SVM	Multilayer Perceptron	75.36
	Radial Basis kernel	87.50
	Linear	87.86
KNN	Correlation	88.21
	Euclidean	88.93
	Cityblock	88.93
	Cosine	90.36

Finally table III compares the mean accuracy of the proposed method with several other methods. It can easily be concluded that using simple KNN classifier, our method provides good mean accuracy to classify left and right hand motor imagery movements than all the methods listed in table III.

VI. CONCLUSION

In this paper, a comprehensive robust method to distinguish left and right hand motor imagery movements offering a promising support for an important application in BCI has been proposed. EEG signals have been successfully classified by extracting mean absolute deviation from the coefficients of auto-correlated EEG signals in the dual tree complex wavelet transform domain. Auto correlation makes the recorded EEG signal stronger as well as the noise is suppressed and as a result the proposed method is more robust. Justification of using MAD as feature has been provided with a number of scatter plots, box plots, histograms and one way ANOVA analysis. Among various kinds of classifiers like

TABLE III. COMPARISON OF MEAN ACCURACY OF DIFFERENT METHODS

Method	Proposed by	Classifier	Mean Accuracy (%)
PSD	Solhjoo et al. [5]	Mahalanobis distance	63.1
		Gaussian classifiers	65.4
		LDA	65.6
Raw EEG	Solhjoo et al. [5]	HMM	77.50
AAR	Tavakolian et al. [7]	Bayes Quadratic	82.86
		BGN	83.57
		MLP	84.29
Morlet wavelet	Lemn et al. [8]	Bayes Quadratic	89.29
Multiple auto correlation	Wang et al. [8]	LVQ	90.00
Proposed method (DTCWT)		KNN	90.36

SVM, KNN, Naïve Bayes classifier and LDA, KNN provides the highest accuracy of 90.36%. Finally the mean accuracy of the proposed scheme has been compared with several other recent methods available in EEG literature and has shown to be superior to the others.

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